

* Computing system consists of

- Processor
- Memory
- I/O

↓

General Purpose Specific Purpose

* Embedded System Constraints: Power, size, speed, cost

* Embedded system methods:

① System Board (SB)

② System on Chip (SOC)

	SB	SOC
Cost	↑	↓ (✓)
Size	↑	↓ (✓)
Performance (Speed)	the same	
Power	↑	↓ (✓)
Configurability	✓	X

* we note that soc is better than SB.

* But in the initial design stages SB is used because it has ability to modify.

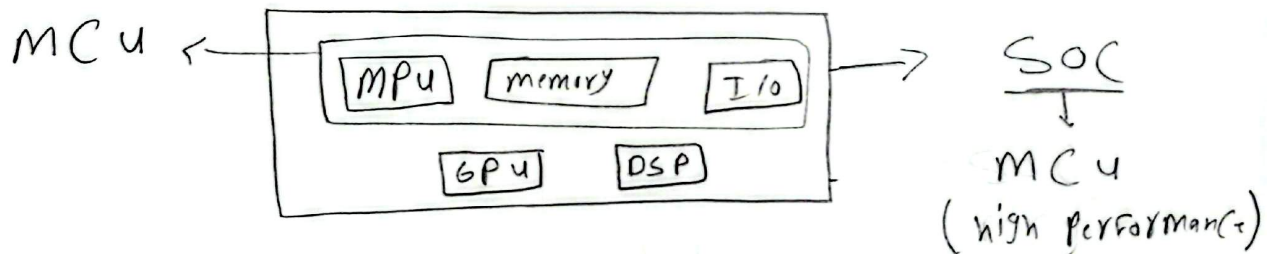
* VLSI → Very large Scale Integrated Circuit.

⑧ MPU: A modern Processor made from tiny transistors, Containing (ALU), a Control unit, Registers.

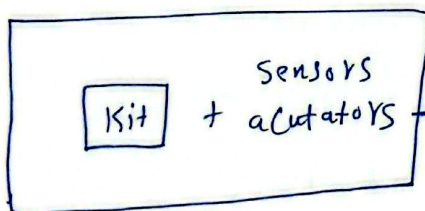
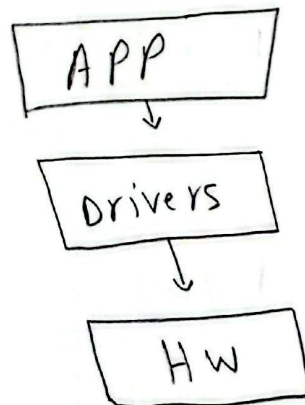
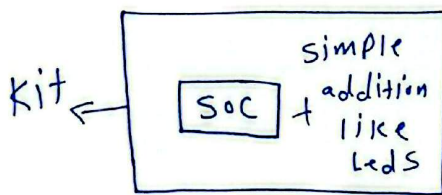
⑧ CPU: The primary Processor in the system.

→ there are secondary processors that assist CPU, like GPU and DSP, to reduce the load on CPU.

⑧ MCU: is a small Computer integrates the three parts of any computing system into a single unit.



⑧ Bare Metal SW:-



ECU

→ The foundational building block for large systems like car.

* Processor:- Consists of

- ALU
- Control unit
- Registers

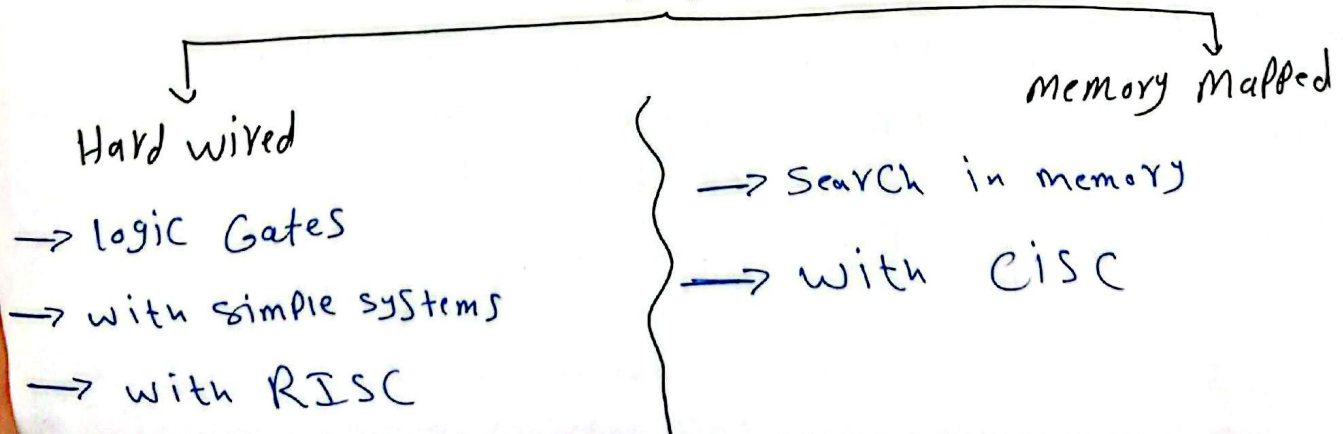
* Instruction Life Cycle:- Every instruction goes through three sequential stages.

* Fetch:- The instruction is brought from (Rom) into Processor. The (PC) Program Counter points to the location of the next instruction, and once fetched, it's stored in an internal register called the Instruction Register (IR).

* Decode:- happens inside Control unit to understand the instructions. This stage depends on Processor's instruction set and instruction format to decode the operation and send the suitable signal to the ALU.

* Execute:- The ALU receives the signal and the operands, performs the required operation, and outputs the result to be stored temporarily in a general purpose register.

ID



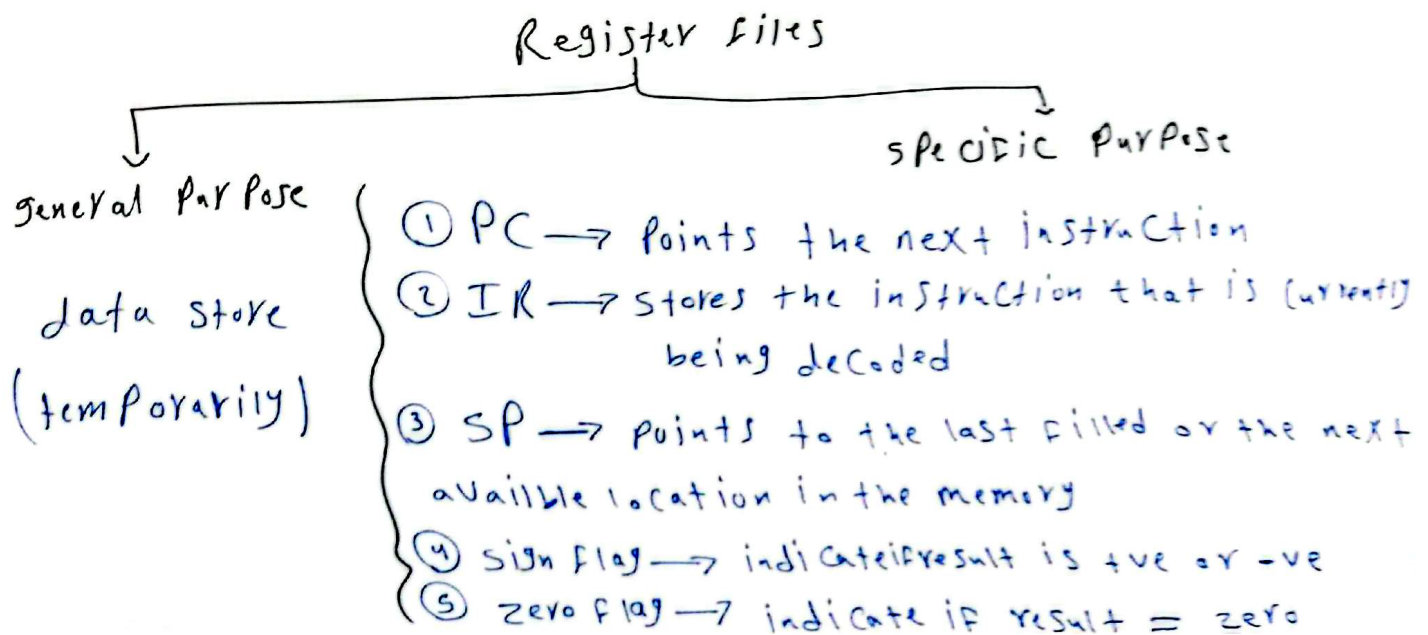
ISA $\rightarrow$ RISC and CISC

Reduced

Complex

ES-Challenges	RISC	CISC
size	ALU $\downarrow$, ID $\uparrow$ $\sim$	ALU $\uparrow$, ID $\downarrow$
Cost	SW $\uparrow$, HW $\downarrow$ $\sim$	SW $\downarrow$, HW $\uparrow$
Performance	same $\sim$	same
Power-Cons.	ALU $\downarrow$, ID $\uparrow$	ALU $\uparrow$, ID $\downarrow$
	Hard-wired	memory-mapped
Compiler	Strong	Simple 6

~~* Rig~~ Register files:-



⊛ Memory Basics:-

memory:- it consists of locations, and each location holds a number of bits. The basic building block of memory is the Flip-Flop.

→ The processor connects to the memory via 3 main buses:
Address Bus, Data Bus, Control Bus.

⊛ Read/write operations:- to read, the processor sends the address on the Address Bus and a (read) signal on the Control Bus, then the memory sends the data via the Data Bus.

→ writing data to memory takes more time than reading from it.

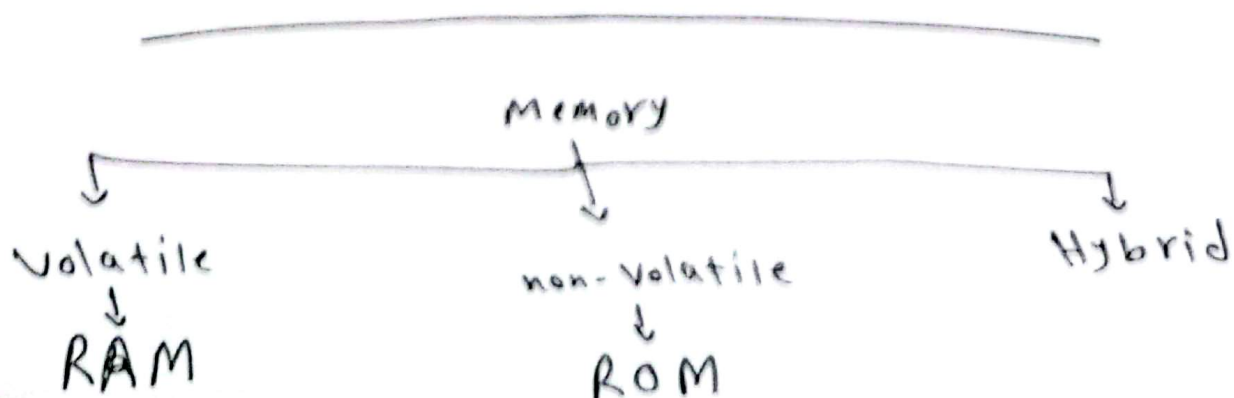
→ The number of (Address Bus) lines determines the maximum size of the memory.

$$\text{⊛ Lines of (Address Bus)} = n$$

$$\text{⊛ memory size} = 2^n$$

⊛ Characteristics for any memory:

① Capacity ② Access time ③ Organization



⊛ Dynamic Ram (DRAM):-

- uses capacitors (1 = charged, 0 = Dis charged)
- The problem: capacitors leak charge over time
- The Solution: Requires a "refresh circuit" to keep data alive
- Slow, but High density and low cost.

⊛ Static RAM (SRAM):-

- uses transistors (at least 6 tran. per bit)
- The problem → high cost and low density.
- the Solution → stable internal loop, doesn't need a refresh circuit
- Very fast compared to DRAM

⊛ How writing to memory works:-

The memory grid consists of word lines and bit lines. to write a specific value to a bit, The ~~bit~~ corresponding word line was activated and required value was entered via the bit line.

* Rom:-

→ CPU → Can only read from it.

* To write on Rom we need external device called a "Burner", "Flasher" or "Flash driver". This device applies a High Voltage to a "Control gate" to break through an insulating layer (SiO_2) and force electrons into a "Floating gate".

* Types of Rom:-

- ① Mask Rom → programmed only once at the Factory.
very Cheap and used for mass Production
- ② PROM → one-Time programmable (OTP) by the user.
It depends on internal fuses, once a fuse is burned to write code, it can't be changed.
- ③ EPROM → Can be erased using ultra violet light through a small window, requires erasing the entire memory at once, and it is sensitive to radiation and noise.